Research CLI — Agent System Architecture

CLI, core session and routing, tools and research modules, model providers, and embedded runtimes

Dashed frame = architecture zone
Solid frame = subsystem
Box = module and responsibility

PROJECT OVERVIEW

Research CLI is a terminal-first AI agent for code, tools, MCP services, and academic research workflows.

The architecture separates the CLI, core session and routing, tools and research modules, model providers, and embedded runtimes.

CLI FRONTEND · packages/cli

- **research.tsx** — interactive TUI
- **commands/** — mcp.ts; skills.tsx · hooks.tsx
- **ui/commands/research** — arxivCommand; paper · investigate · submit
- **acpSession.ts** — ACP session; stdio transport

CORE ORCHESTRATION · packages/core

Session

- **researchChat.ts** — researchRequest.ts
- **research-config-manager.ts** — academic settings
- **config / memory** — context window

Routing

- **modelRouterService** — routing/modelRouterService.ts
- **defaultStrategy** — routingStrategy.ts
- **classifierStrategy** — gemma · numerical
- **compositeStrategy** — approval · fallback; override

TOOLS & RESEARCH MODULES

Workspace

- **edit.ts · grep.ts** — filesystem changes
- **shell / ls** — local execution

MCP

- **mcp-client.ts** — server sessions
- **mcp-client-manager.ts** — resource routing

analysis/

- **bibliography-manager** — citations
- **experiment-code-generator** — Python · R · Julia
- **research-data-analyzer** — statistics · ML

bibliography/

- **arxiv-mcp-client** — arXiv search
- **google-scholar-client** — scholar search
- **literature-investigator** — keyword-generator; reference-classifier

submission/

- **latex-manager** — project compile
- **journal-matcher** — venue selection
- **arxiv-latex-extractor** — template extract

MODEL PROVIDERS

- **Gemini / Vertex**
- **OpenAI**
- **Anthropic**
- **Baidu ERNIE**
- **DeepSeek**
- **Qwen**
- **Groq / Mistral** — Cohere · Moonshot

EMBEDDED RUNTIMES

- **packages/sdk** — library embedding
- **packages/a2a-server** — agent executor; HTTP task API
- **vscode-ide-companion** — IDE companion
- **packages/devtools** — development tools

The CLI and ACP form the frontend. Core owns session and model routing. Tools, research modules, providers, and embedded runtimes are separate boundaries.